

Homework_8

Integrative Genomics Viewer (IGV): Overview and Analytical Utility

To facilitate integrative, locus-specific analysis of gene regulation, we utilized the Integrative Genomics Viewer (IGV), a high-performance visualization platform developed by the Broad Institute. IGV enables interactive exploration of large-scale genomic data and is optimized for next-generation sequencing (NGS) datasets, including RNA-seq, ChIP-seq, DNA-seq, and epigenetic profiles. Its core functionality includes real-time navigation of aligned reads, gene models, variant calls, and coverage tracks at both genome-wide and nucleotide-level resolutions.

IGV's strength lies in its ability to display multiple synchronized data tracks, allowing users to visually correlate transcription factor binding (e.g., ChIP-seq) with transcriptional activity (e.g., RNA-seq) across genomic loci. For this study, IGV was employed to compare gene expression and RNA Polymerase II (POLR2A) occupancy between two human cell lines—K562 and MCF7—using the hg19 genome build as reference.

Assignment Workflow and Data Integration

To perform this visualization, the following data types were uploaded into IGV for each cell line:

- **RNA-seq data** (BAM format), containing uniquely mapped reads to evaluate transcript expression levels.
- **POLR2A ChIP-seq data** (BAM format), representing uniquely aligned reads to identify POLR2A binding sites.
- **POLR2A peak files** (BED format, narrowPeak), summarizing ChIP-seq enrichment regions.
- **RNA-seq-derived mutation calls** (VCF format), identifying transcriptomic variants for exploratory interpretation.

Once all tracks were loaded, five genomic loci were analyzed corresponding to the following genes: **HBG1**, **RAD21**, **CREB1**, **POLR2A**, and **SMC2**. Using IGV's coordinate-based search functionality, we manually navigated to each locus and examined:

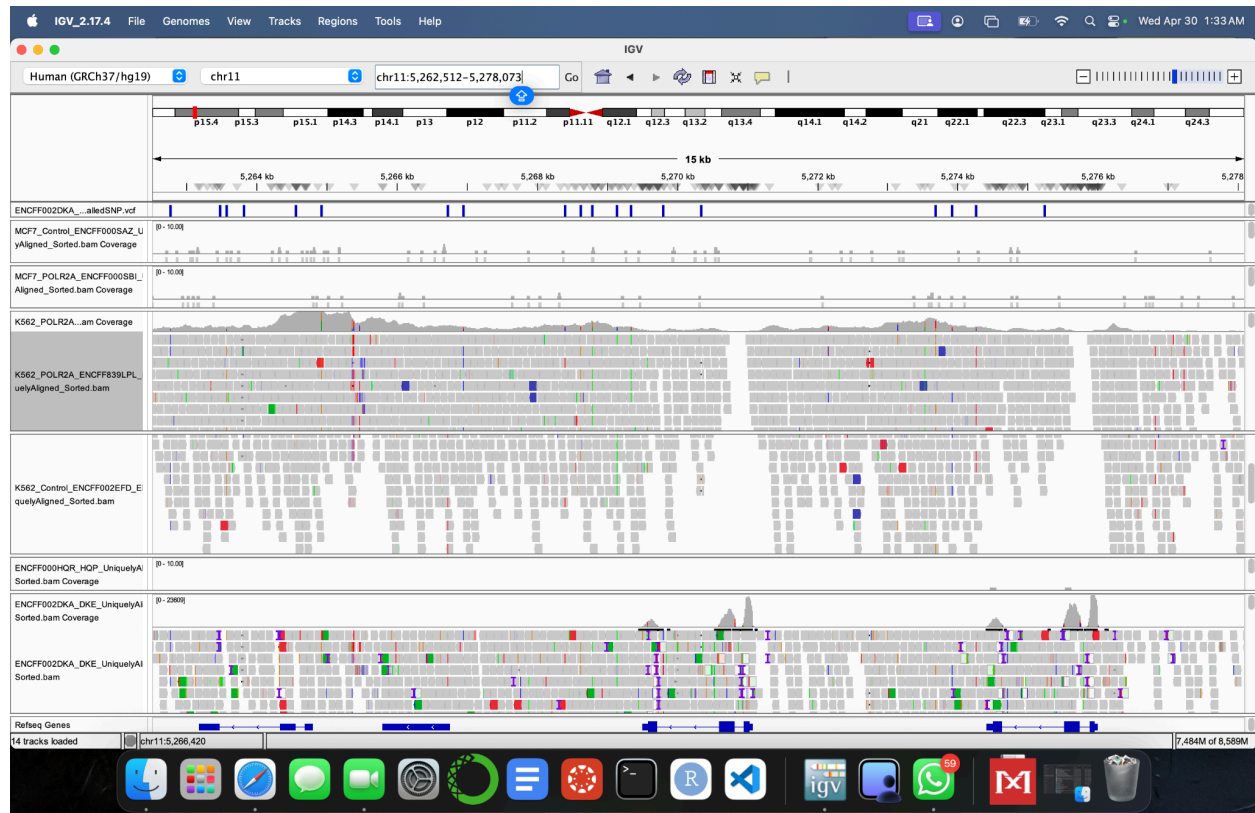
- The presence and distribution of POLR2A binding peaks from ChIP-seq data.
- The RNA-seq coverage plots, indicating gene expression levels.
- Comparative differences in signal intensity and read density between K562 and MCF7.
- Read alignment details, including splicing and any mutations indicated in the VCF tracks.

No additional statistical processing or computational modeling was performed outside of IGV. The analysis was fully visual and exploratory in nature, focused on interpreting regulatory and transcriptional differences between the two cell types at a high resolution. This integrative

approach provided a qualitative understanding of gene-specific transcriptional activity in relation to POLR2A occupancy.

Below is a locus-by-locus breakdown of our observations:

HBG1



Summary

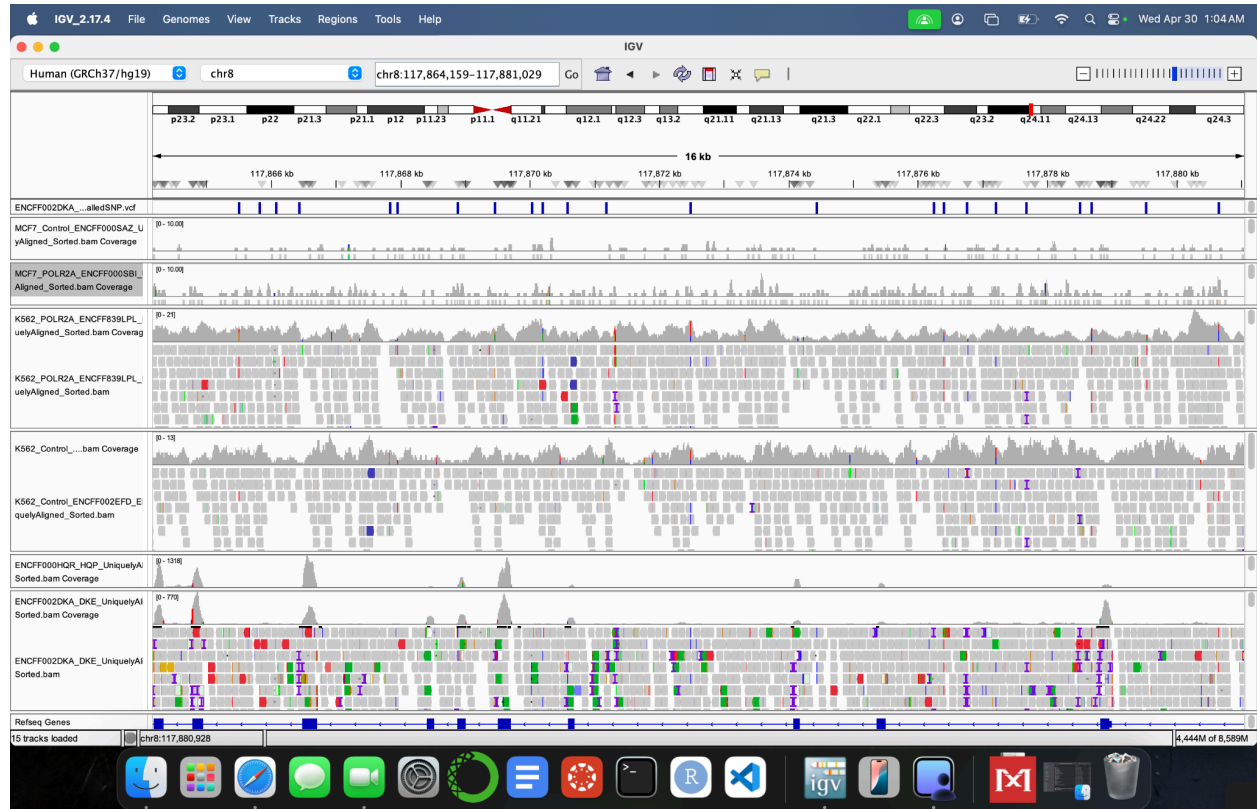
1. HBG1 (chr11:5,262,512–5,278,073)

In K562 cells, a strong POLR2A binding peak was observed across the HBG1 gene body, suggesting active transcription. The RNA-seq track confirmed high expression, with dense coverage and well-aligned reads. In contrast, MCF7 cells showed no detectable POLR2A enrichment at the HBG1 locus, and RNA-seq data revealed negligible expression.

These findings are consistent with the hematopoietic origin of K562, where HBG1 (a fetal hemoglobin gene) is developmentally regulated and actively transcribed. In epithelial-derived MCF7, where globin genes are typically inactive, both POLR2A binding and transcription are absent.

This example demonstrates a strong correlation between POLR2A occupancy and active transcription, specifically in a lineage-dependent manner.

RAD21



Summary

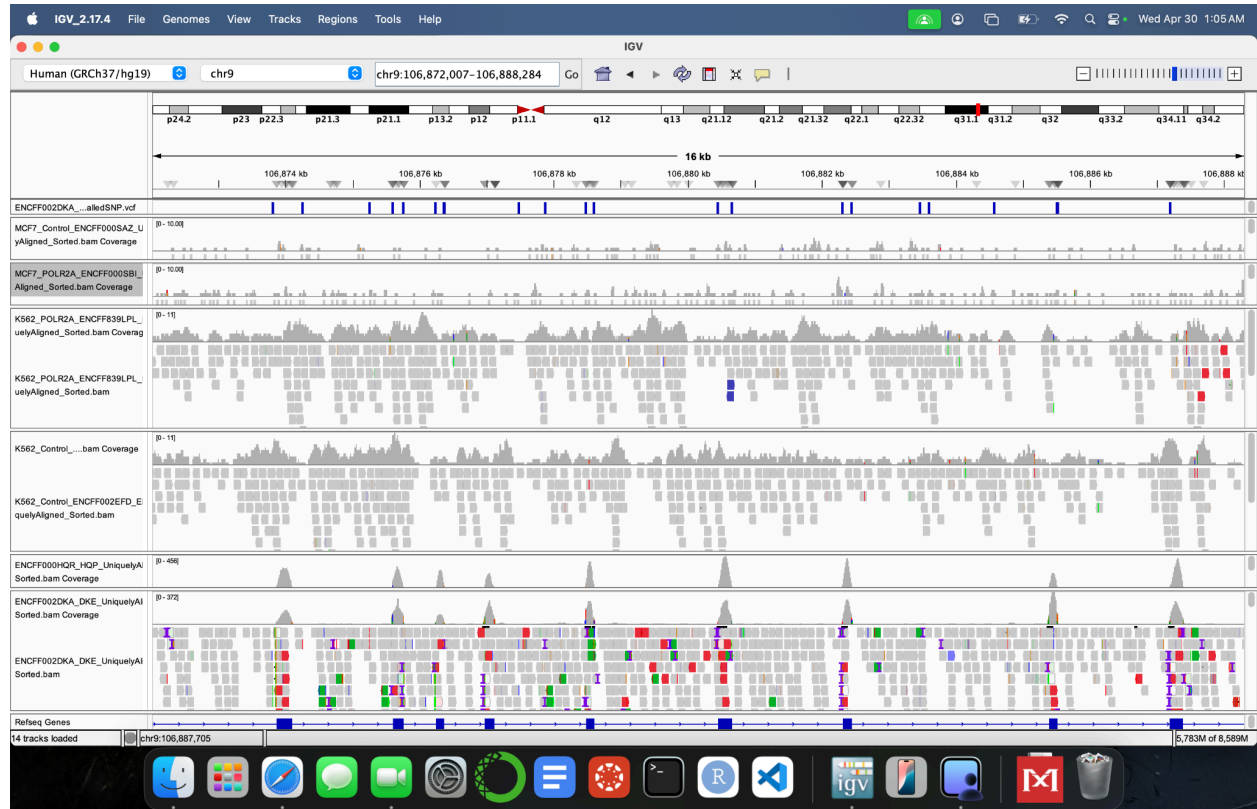
RAD21 (chr8:117,864,159–117,881,029)

Both cell lines exhibit POLR2A binding at the RAD21 locus, although MCF7 shows stronger and broader peaks compared to K562. RNA-seq data similarly reveal higher expression in MCF7 and moderate expression in K562, reflecting the POLR2A signal intensity.

RAD21 encodes a cohesin complex subunit involved in chromatin organization, and its higher transcriptional activity in MCF7 may reflect differences in regulatory demands between these two cancer types.

The concordance between POLR2A enrichment and RNA-seq coverage supports the role of POLR2A in active transcription and indicates transcriptional upregulation of RAD21 in MCF7.

SMC2



Summary

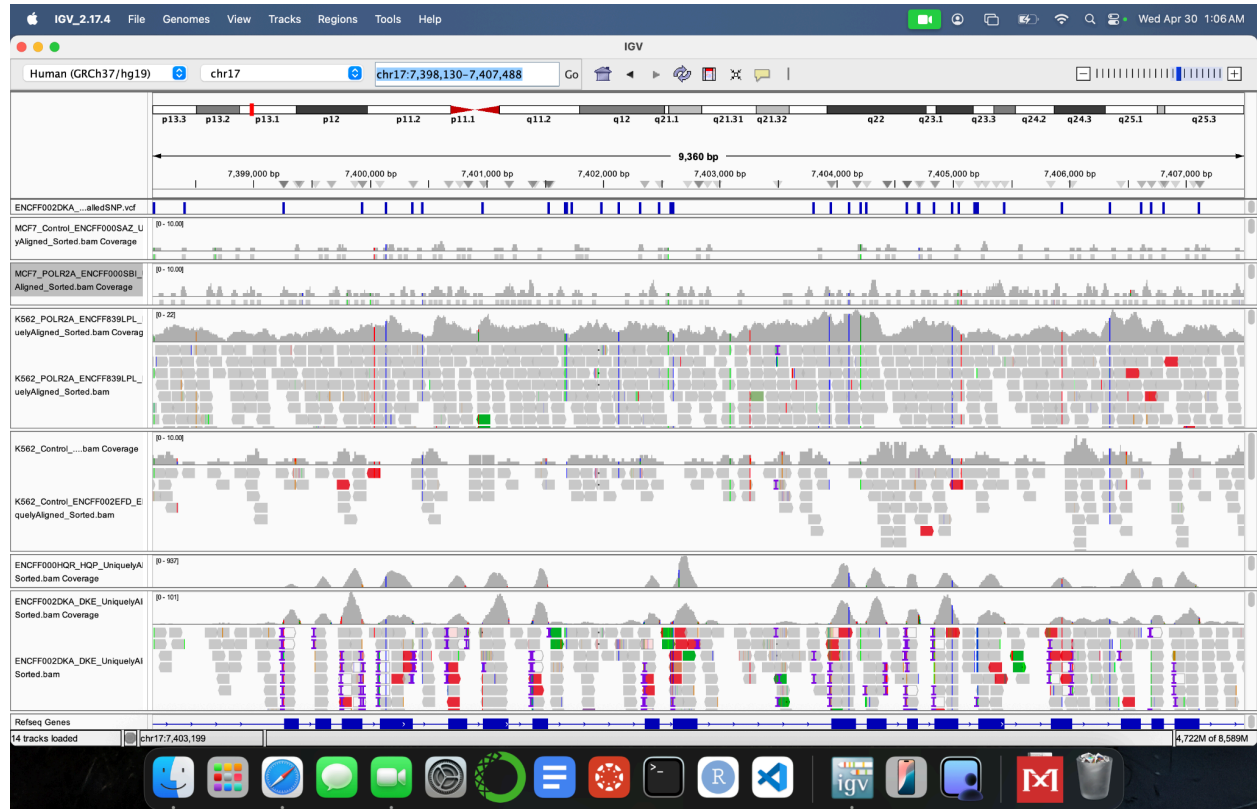
SMC2 (chr9:106,872,007–106,888,284)

In K562, moderate POLR2A peaks are observed at the SMC2 locus, consistent with moderate gene expression shown by RNA-seq coverage. In contrast, MCF7 cells exhibit little to no POLR2A binding, and the corresponding RNA-seq track indicates minimal or absent transcription.

SMC2 encodes a condensin complex subunit, and its expression pattern suggests cell-type-specific transcriptional regulation, potentially linked to chromosomal architecture differences between hematopoietic and epithelial cells.

This supports the idea that POLR2A binding can serve as a reliable proxy for transcriptional activity, especially in genes with variable expression across cell types.

POLR2A



Summary

POLR2A (chr17:7,398,130–7,407,488)

Unsurprisingly, both cell types show extensive and strong POLR2A enrichment across the POLR2A gene itself. RNA-seq tracks indicate very high expression levels in both K562 and MCF7.

As the gene encoding the largest subunit of RNA Polymerase II, POLR2A is a housekeeping gene that is ubiquitously and constitutively expressed. The self-enrichment of POLR2A at its own locus may reflect autoregulatory mechanisms or sustained polymerase occupancy due to high transcriptional demand.

This locus served as a positive internal control, confirming the integrity and reliability of the ChIP-seq and RNA-seq datasets.

CREB1



Summary

CREB1 (chr2:208,418,683–208,446,715)

The CREB1 locus exhibits broad POLR2A binding across the gene body in both K562 and MCF7 cells. RNA-seq coverage confirms moderate expression levels in both cell types. As a transcription factor involved in the cAMP response element-binding pathway, CREB1 is widely expressed and participates in a range of cellular processes. The ChIP-seq and RNA-seq data demonstrate that CREB1 is constitutively expressed in both lines, with POLR2A binding indicative of ongoing transcription. This reinforces the notion that stable POLR2A engagement across a gene body correlates with continuous transcriptional activity.

General Observations and Conclusions

Across all five loci examined, POLR2A ChIP-seq enrichment patterns were closely aligned with RNA-seq-derived expression levels, reinforcing the role of POLR2A as a reliable marker of active transcription. HBG1 and SMC2 were found to be transcriptionally active only in K562, with strong POLR2A occupancy, indicating lineage-specific gene expression characteristic of the myeloid lineage. In contrast, RAD21 and CREB1 were expressed in both K562 and MCF7 cells, although RAD21 exhibited notably higher transcriptional activity in MCF7, as reflected by increased POLR2A binding and RNA-seq coverage. As expected, POLR2A itself, being a core component of the transcriptional machinery, showed uniformly high expression and strong binding signals in both cell lines, serving as an internal control for data quality and alignment accuracy.